

Linear:

Linear regression model performs strongly and shows that previous performance is the primary predictor, while study hours and motivation are important actionable factors that relate positively to student grade outcomes.

Evaluation for linear part:

On the test set, the model achieved $R^2 \approx 0.877$ and $MAE \approx 3.98$, meaning it explains most of the variation in 'final_score' and typically predicts within about 4 points of the true score. This supports using regression as a reliable, interpretable way to study influential factors. The top-5 coefficients are all positive, which aligns with intuition that stronger background knowledge, more effort and higher motivation will help a student achieve higher final score.

Logistic:

Logistic regression model produces high-quality pass/fail predictions. The key drivers of success are dominated by prior performance, while attendance, study hours, and motivation provide important, actionable signals that can be targeted to improve student outcomes.

Evaluation for logistic part:

On the test set, it achieved Accuracy = 0.95, F1 = 0.973, and ROC-AUC = 0.978. This indicates the model can separate passing vs. failing students with very strong reliability. The confusion matrix shows the model makes very few mistakes overall, and it is especially strong at identifying students who will pass (183 correct predictions).

The top influential factors are led by 'math_prev_score' and 'previous_score', both with large positive coefficients. This means students with stronger prior performance have a substantially higher probability of passing. Beyond prior scores, 'motivation_score', 'attendance_percentage', and 'daily_study_hours' are among the top drivers and all contribute positively. This supports that by improving motivation, attendance, and study time can meaningfully increase success likelihood.

Lasso:

Lasso regression confirms the same main story as linear regression which prior academic performance dominates, while daily study hours and motivation are important actionable predictors. It also provides a cleaner, more interpretable model by selecting a small set of influential features without sacrificing accuracy.

Evaluation for Lasso part:

On the test set, Lasso reached $R^2 \approx 0.879$ and $MAE \approx 3.95$, which is essentially as strong as the linear regression baseline. This shows that the relationship between features and 'final_score' can be captured well even with regularization and feature selection.

Lasso kept 12 non-zero features, meaning it automatically removed many redundant predictors by shrinking their coefficients to zero. This is useful for interpretability and supports the claim that only a small subset of variables is needed to explain most of the variation in final grades. The selected $\alpha \approx 0.109$ indicates a moderate level of regularization. This makes the final feature set more stable and less sensitive to noise.

Visualization results:

Linear and Lasso give almost identical top-5 and very similar coefficient sizes. It means the influential factors are stable, not sensitive to the modeling choice. In both regressions, 'previous_score' is clearly the largest bar, followed by 'math_prev_score' and 'language_prev_score'. In Logistic regression, 'math_prev_score' and 'previous_score' are again the top two. So the dataset is telling a consistent story that past achievement is the strongest predictor. In addition, 'daily_study_hours' and 'motivation_score' show up in all three plots with positive coefficients. This supports a practical conclusion that even after accounting for prior academic strength, effort and motivation also influences a student's academic performance a lot.

Correlation Analysis:

The correlation heatmap shows strong positive relationships among prior score variables, indicating multicollinearity. This explains why Lasso regularization is useful for shrinking redundant predictors. Behavioral factors such as study hours, attendance, and motivation show moderate positive correlations with final_score, suggesting independent contributions to performance.

Model Without Prior Academic Performance:

After removing prior score variables, linear regression achieved $R^2 \approx 0.71$ and $MAE \approx 6.09$. Although performance dropped compared to the full model ($R^2 \approx 0.88$), the model still explains a substantial portion of grade variation. This confirms that non-academic factors provide meaningful predictive information beyond past achievement.

Actionable Factors Only Model:

Using only controllable behavioral variables, the model achieved $R^2 \approx 0.72$ and $MAE \approx 5.99$. Even without prior academic history, actionable factors alone retain strong explanatory power. This highlights that study habits, attendance, and motivation meaningfully influence student outcomes.

Overall Conclusion:

Prior performance is the strongest predictor, but behavioral and lifestyle factors remain significant and practically important. These results support using actionable variables for future intervention and recommendation systems.